BLWP Waterway Contamination Progress Report 2

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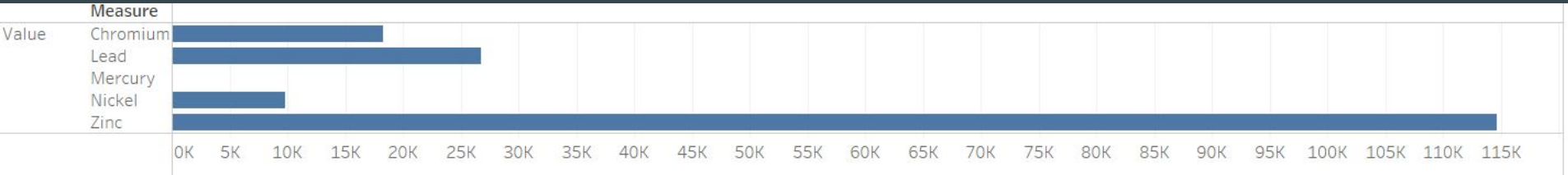
Contributions

- Dylan Long: Wrote first paragraph on on slide 3 discussing errors in collection (slide 3)
- Dylan Long: Explored inconsistent measurements taken from 1998-2016 in each location with a graph and written analysis (slide 8)
- Dylan Long: Explored mercury contamination and potential river flow pattern through graph and written analysis (slide 9)
- Dylan Long: Explored lead contamination and potential river flow pattern and contamination travel through graph and written analysis (slide 10)
- Evan Marino: Looked into inconsistencies of how many unique chemicals are being searched for per year (slide 4)
- Evan Marino: Looked into inconsistencies of what day of the month the measurements are being taken (slide 9)
- Luke Dasilva: created graph on (slide 7), talked about why the data is inconsistent.
- Luke Dasilva: Changes that will prevent waterways to have more testing than others (slide 8)
- Luke Dasilva: added ways that can fix the inconsistent testing on (slide 6)
- Santiago Alvarez: Created Graphs to show a similar consistency on the data taken in the years (slide 13-14)
- Santiago Alvarez: Created a visual on the quantity of harmful elements on the selected locations(slide 15-16)
- Santiago Alvarez: Created a visual to show all of locations with harmful elements and it hints to the locations with waste dump(slide 16)
- Alex Schifferle: Explored the problems with some of the elements found within the water (slide 3)
- Alex Schifferle: Looked into the readings by location to determine which areas had the most contamination (slide 4)
- Alex Schifferle: Looked at the map for clues as to why certain areas were worse than others (slide 5)

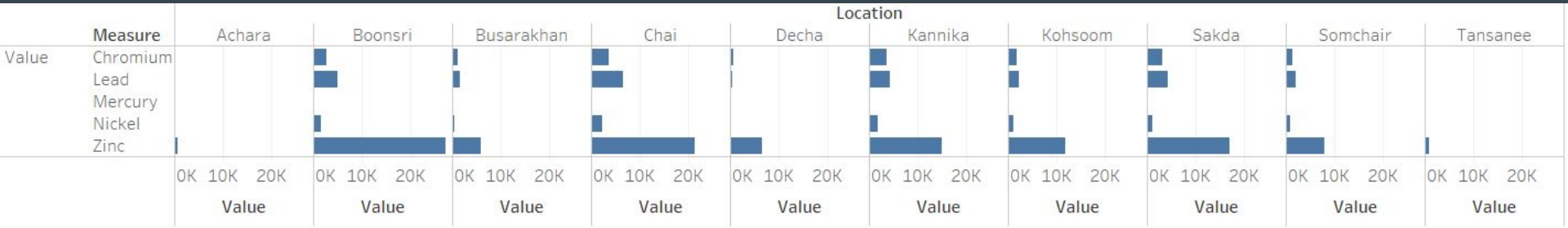
Issues with the water

- Lead, causes a multitude of problems such as because it's highly toxic. Lead can build up in tissue and cause many problems, including death.
- Chromium, causes lots of issues such as organ damage, behavioral change/problems and more.
- Mercury, this is a highly toxic substance. This can cause problems with development, reproductive system and growth.
- Zinc, causes physical and psychological damage. The more zinc concentrations the worse.
- Nickel, this can cause serious impairment to needed functions respiration and ion regulation leading to more issues.

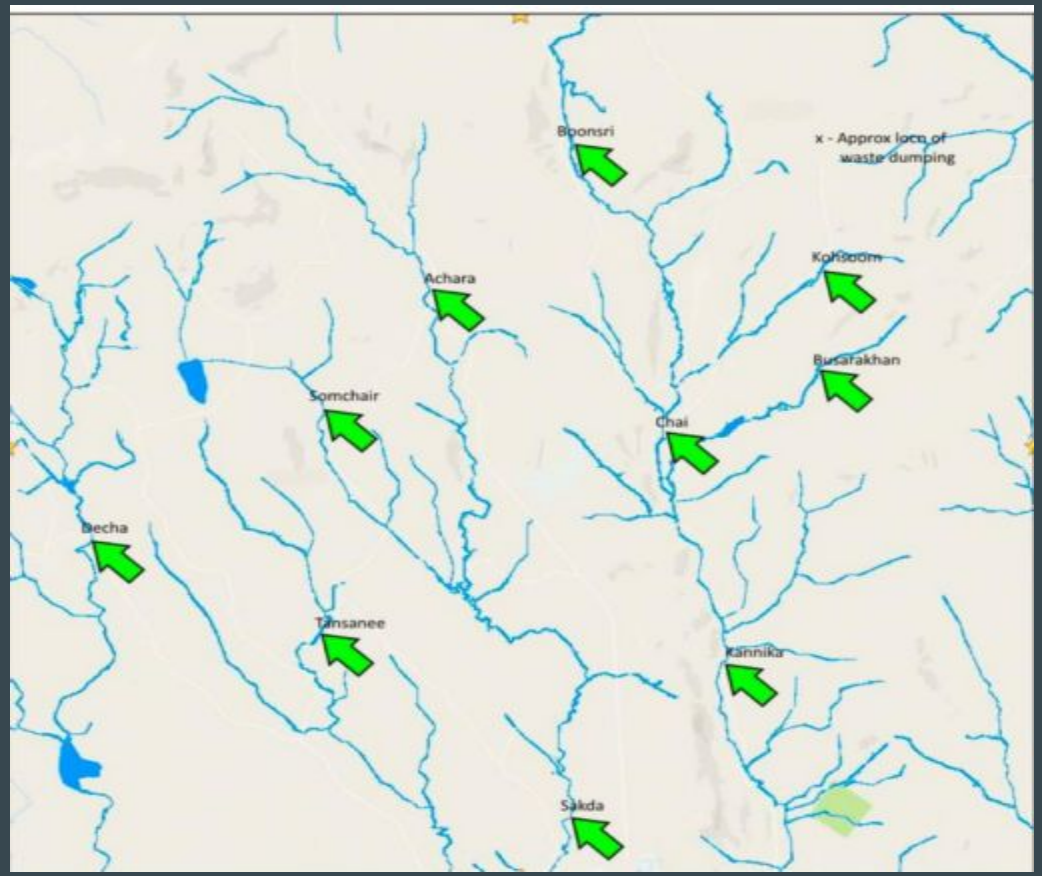


Readings By Location

- Boonsri and Chai have the largest readings of harmful elements
- Then followed by Kannika, Sakda and Kohsoom
- Somchair being in the middle of readings
- Achara, Decha and Tansanee almost have none

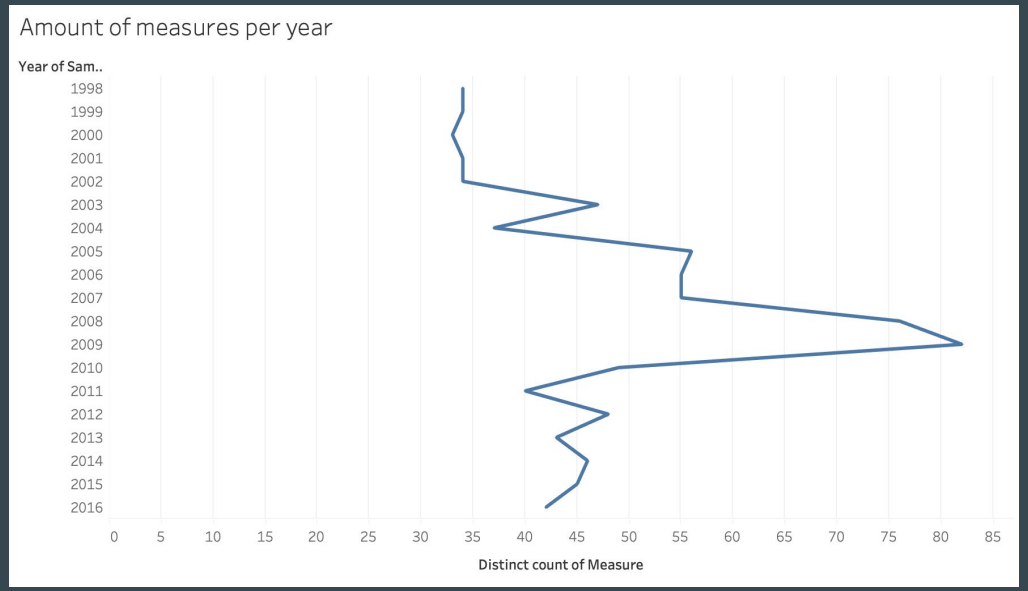


Paths of the water. Boonsri and Kohsoom being the two closest. Water will follow the path of of the waterways. The locations with the least amount of harmful contaminants are further.



Errors in Collection

- Though Boonsri and Chai appear to have high measurements, the amount of measurements recorded at these two locations is overwhelmingly more consistent than the other locations.
- The amount of unique chemical measurements per year also varies highly, with a difference of 82 and 33 between 2009 and 2000



Count of readings in each water way per year

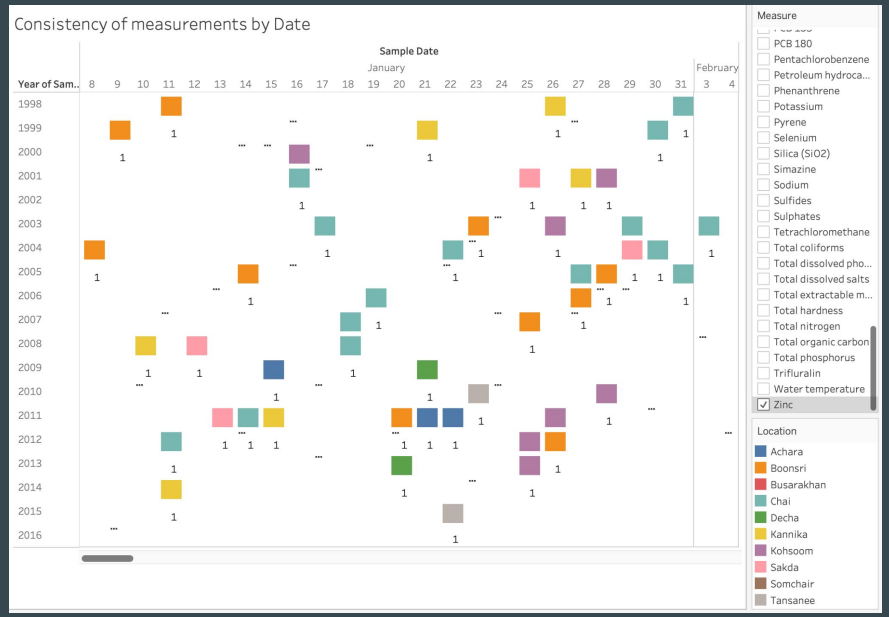
Location	Sample Date																		
	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Achara												321	416	483	426	296	285	314	314
Boonsri	680	829	1,049	818	1,310	1,481	1,898	3,267	3,172	2,704	2,342	2,311	1,403	1,117	1,061	1,448	1,836	1,287	1,301
Busarakhan	96	347	371	300	297	315	295	514	514	627	482	532	391	323	351	377	463	430	467
Chai	1,170	1,029	1,088	912	1,489	1,691	1,569	2,644	2,668	3,045	1,811	2,200	1,718	1,458	1,143	1,193	1,396	1,312	1,709
Decha												282	283	286	351	402	400	365	362
Kannika	1,164	1,003	1,091	977	843	895	738	1,490	1,559	1,882	1,301	1,624	1,212	1,224	747	737	1,125	1,263	1,277
Kohsoom	196	334	376	335	301	353	349	523	527	570	533	554	462	425	413	469	424	371	380
Sakda	393	1,041	1,091	986	837	870	860	1,494	1,528	1,882	1,428	1,602	1,212	1,158	738	731	1,107	1,259	1,212
Somchair	194	343	370	293	260	314	276	516	525	628	476	550	391	325	335	381	442	440	478
Tansanee												244	337	238	233	216	292	334	280

- This graph shows each waterway and what year had the most readings of chemicals
- This shows from the years 2005 to 2009 has the most readings, which supports that there are inconsistent data readings throughout the years.
- Also showing that in Achara, Decha and Tansanee, there were no reading until the year 2009.

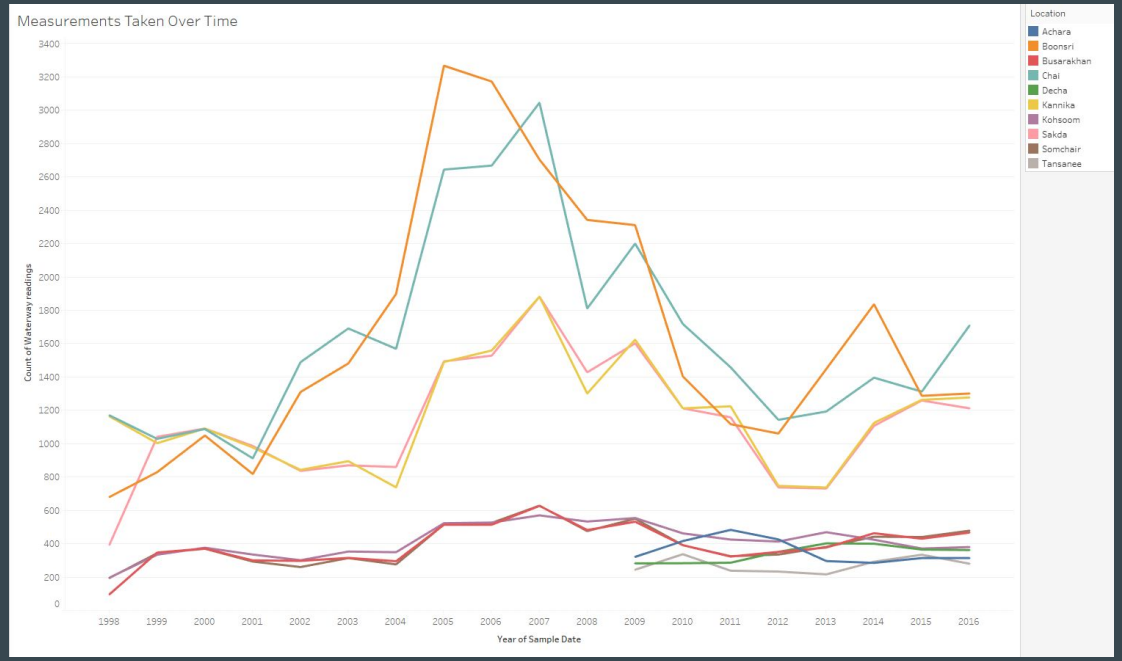
Changes that could make the approach easy to understand:

- Increase the sampling frequency at all sights, and focus even more on locations that are under-sampled (some sites are sampled quarterly or annually, make each location the same and consistent)
- Focus more on the areas near the waste dumping, like Boonsri, Kohsoom, and Busarakhan, these waterways have the most contamination and spread throughout to the other waterways. A main focus on these waterways will help keep the others clean.
- Additional sampling stations at key downstream points near the contamination sources and ensure all chemicals are sampled the same across all locations
- Introduce event-based sampling for weather conditions, Find where heavy rain is, droughts or industrial activity.

- Make a schedule for the whole Preserve to follow for what days each month to hold the readings so that the differences in the values at different locations are more valuable
- Limit redundancy with the amount of chemicals being measured to focus on the ones that can be harmful if the value increases too much
- Ensuring values like Zinc, Mercury, and Lead are being measured is more important than having a wider range of unique measurements that have gaps of not being measured

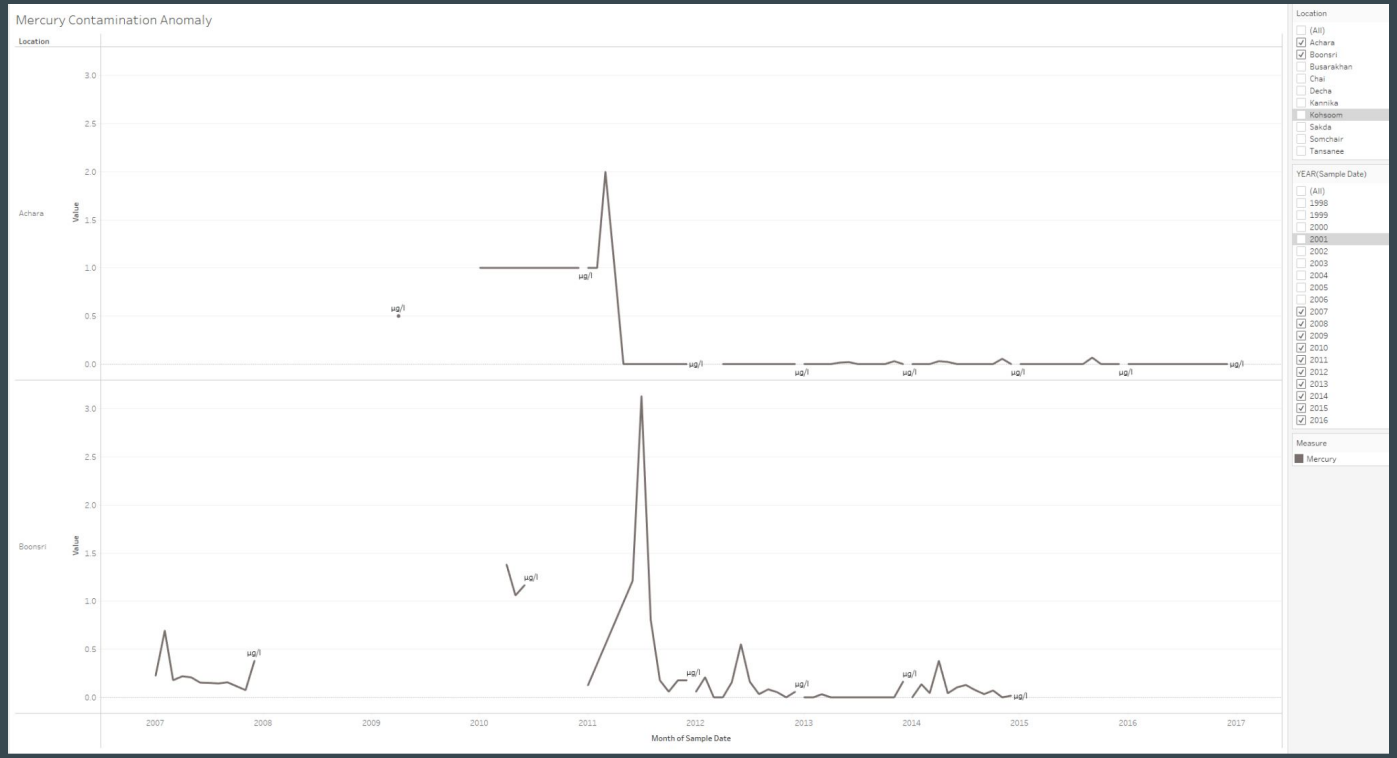


Measurements Over Time



- Ideally, we would want this graph to be a straight line across.
- The peaks indicate a large amount of recordings taken in the location per year.
- Notice that Somchair, Kohsoom, and Busurakhan are all relatively consistent.
- Notice that measurements in Tansanee, Decha, and Achara did not even start until 2009.

Mercury Contamination Anomaly

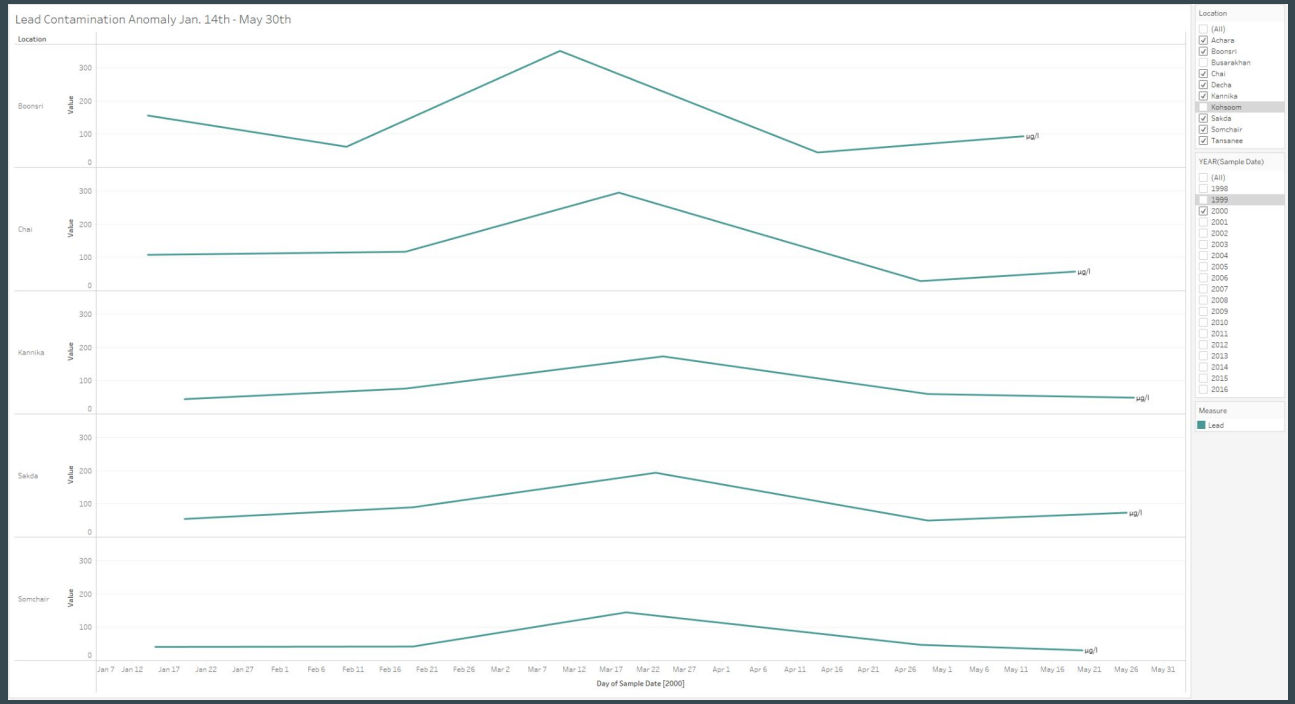


In March 2011, there was an anomalous spike of mercury in Achara.

In June of 2011, a large measurement was recorded in Boonsri while the measurement in Achara decreased.

This indicates a possible flow pattern that goes from Achara towards Boonsri.

Lead Contamination Anomaly



- Note the rapid increase starting in Boonsri on Feb 10th.
- Boonsri lead levels peak on March 10th and slowly decline while peaking sequentially at downriver locations according to the provided map.
- This approximates that the water flows south from Boonsri and that the lead contamination originates within close proximity of the suspected dumping location.

Patterns of Contamination

Findings:

- **Boonsri:** Shows consistent pattern similarities in data when samples are taken downstream from the water flow.
- **Chai:** Demonstrates a similar pattern to Boonsri, indicating comparable environmental factors.
- **Kannika:** Also aligns with Boonsri's monthly sample trends, suggesting regional consistency.

Conclusion:
The similar patterns in most locations, except for Busarakhan, suggest the influence of a nearby event, possibly near the waste dump site, impacting water flow and sample characteristics. The spikes observed at Sakda could be due to an underwater passage, carrying contaminants into that area and affecting the water quality

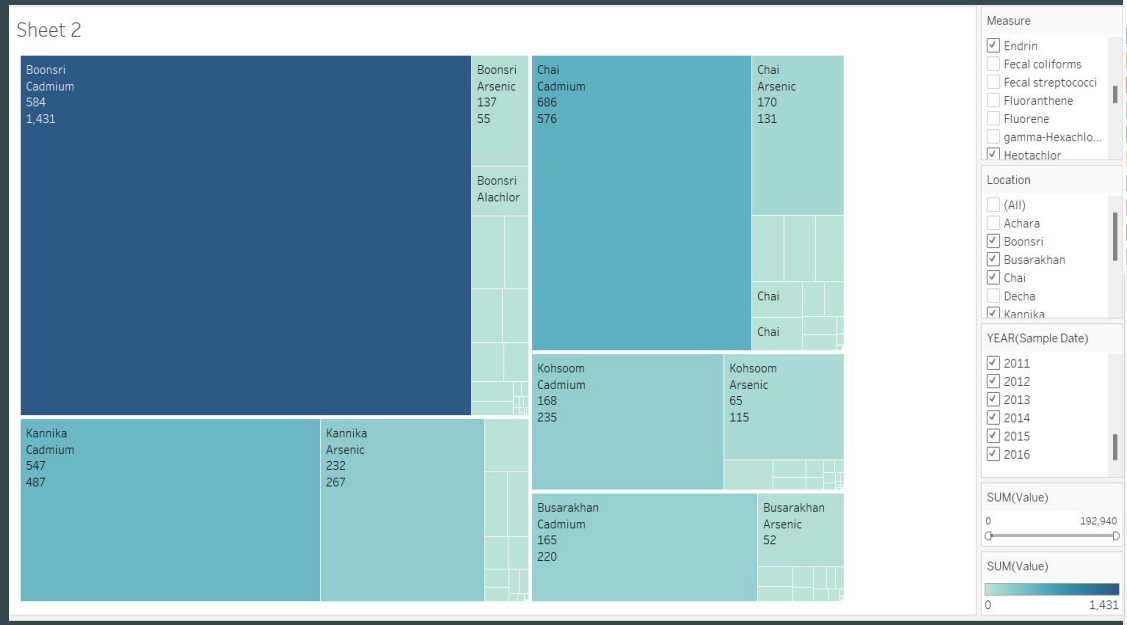
Sheet 4
Location / Sample Date
Boonsri
Chai
Kannika
Kohsoom
Busarakhan
Sakda
Value
80K
60K
40K
20K
0K
Count of in
2K
1K
0K
-1K
YEAR(Sample Date)
2005
2006
2007
2008
2009
2010
2011
Location
Busarakhan
Chai
Decha
Kannika
Kohsoom
Sakda
Somchair
MONTH(Sample Date)
(All)
January
February
March
April
May
June
Location
Boonsri
Chai
Kannika
Kohsoom
Busarakhan
Sakda

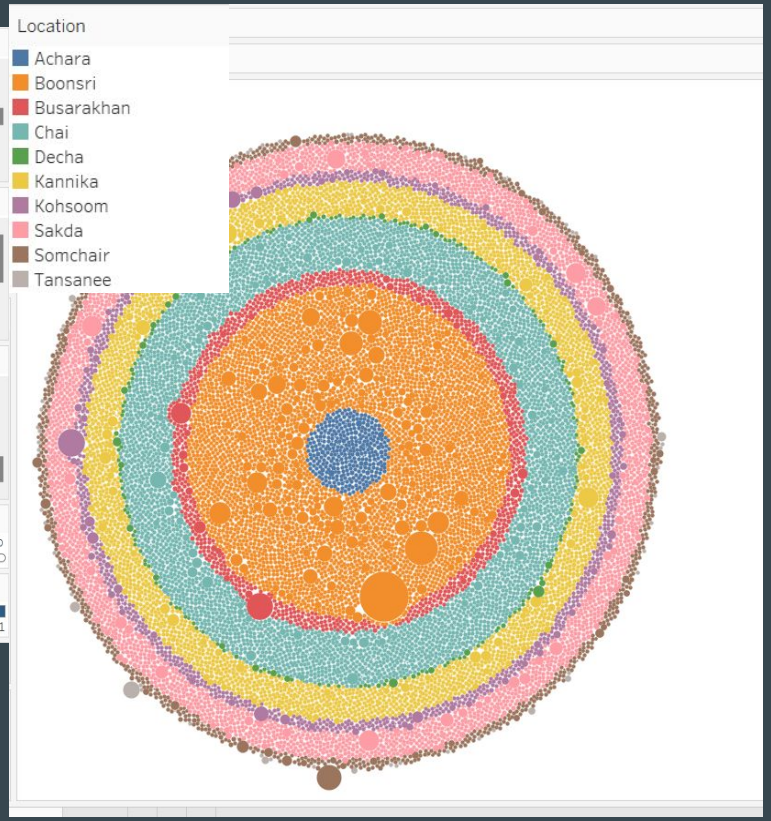
Toxic Elements in Different Locations

This treemap visualizes the concentration of several toxic elements—Cadmium, Arsenic, and Alachlor—in locations such as Boonsri, Chai, and Kannika. The darker and larger rectangles represent higher concentrations of each element.

Key Toxic Elements Contributing to Water Contamination:

1. **Cadmium** - Interferes with fish reproduction and weakens immune systems.
2. **Arsenic** - Damages gills, accumulates in aquatic life, and is toxic at high levels.
3. **Alachlor** - Harms aquatic plants and disrupts fish growth.
4. **Fluoranthene** - Carcinogenic, affects fish liver function and increases mortality.
5. **Heptachlor** - Damages nervous systems in fish and other aquatic species.
6. **Endrin** - Extremely toxic, causing paralysis and mortality in aquatic animals.


Sheet 2
Boonsri
Cadmium
584
1,431
Boonsri
Arsenic
137
55
Boonsri
Alachlor
Chai
Cadmium
686
576
Chai
Arsenic
170
131
Chai
Chai
Kannika
Cadmium
547
487
Kannika
Arsenic
232
267
Kohsoom
Cadmium
168
235
Kohsoom
Arsenic
65
115
Busarakhan
Cadmium
165
220
Busarakhan
Arsenic
52
Measure
Endrin
Fecal coliforms
Fecal streptococci
Fluoranthene
Fluorene
gamma-Hexachlo...
Heptachlor
Location
(All)
Achara
Boonsri
Busarakhan
Chai
Decha
Kannika
YEAR(Sample Date)
2011
2012
2013
2014
2015
2016
SUM(Value)
0
192,940
SUM(Value)
0
1,431


Location
Achara
Boonsri
Busarakhan
Chai
Decha
Kannika
Kohsoom
Sakda
Somchair
Tansanee